UNIVERSIDAD TECNOLÓGICA DE CHIHUAHUA

ING. TECNOLOGÍAS DE LA INFORMACIÓN EN EI ÁREA DE DESARROLLO DE SOFTWARE

9N Desarrollo Web Integral

Documento de Reporte: [Práctica - 1] Pruebas Unitarias

IDGS91N

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1. Configuración del Entorno y Dependencias (pom.xml)

Inclusión de la dependencia spring-boot-starter-test en el archivo de configuración de Maven para dar soporte a JUnit 5 y Mockito.

Captura / Código de la configuración en pom.xml*:*



```
mx.edu.utch.barrosoya10.alumno   pom.xml 1
pom.xml
<project xmlns="http://maven.apache.org/POM/4.0.0"
    <build>
        <plugins>
            <plugin>
                <groupId>org.springframework.boot</groupId>
                <artifactId>spring-boot-maven-plugin</artifactId>
            </plugin>
            <plugin>
                <groupId>org.jacoco</groupId>
                <artifactId>jacoco-maven-plugin</artifactId>
                <version>${jacoco.version}</version>
                <executions>
                    <execution>
                        <goals>
                            <goal>prepare-agent</goal>
                        </goals>
                    </execution>
                    <execution>
                        <id>generate-coverage-report</id>
                        <phase>test</phase>
                        <goals>
                            <goal>report</goal>
                        </goals>
                    </execution>
                    <execution>
                        <id>enforce-service-coverage</id>
                        <goals>
                            <goal>check</goal>
                        </goals>
                        <configuration>
                            <rules>
                                <rule>
                                    <element>CLASS</element>
                                    <includes>
                                        <include>mx.edu.utch.barrosoya10.alumno.AlumnoService</include>
                                    </includes>
                                    <limits>
                                        <limit>
                                            <counter>LINE</counter>
                                            <value>COVEREDRATIO</value>
                                            <minimum>0.90</minimum>
                                        </limit>
                                        <limit>
                                            <counter>BRANCH</counter>
                                            <value>COVEREDRATIO</value>
                                            <minimum>0.90</minimum>
                                        </limit>
                                    </limits>
                                </rule>
                            </rules>
                        </configuration>
                    </execution>
                </executions>
```

2. Implementación de la Clase de Pruebas Unitarias (*Test.java)

Creación de la clase de pruebas aisladas utilizando las anotaciones @Mock e @InjectMocks para simular el comportamiento del repositorio y validar la lógica del servicio.

AlumnoServiceTest.java

src > test > java > mx > edu > utch > barrosoya10 > alumno > AlumnoServiceTest.java > { } mx.edu.utch.barrosoya10.alumno

```java
@ExtendWith(MockitoExtension.class)
class AlumnoServiceTest {

    @Mock
    private AlumnoRepository alumnoRepository;

    @InjectMocks
    private AlumnoService alumnoService;

    @Test
    void findAllReturnsRepositoryResults() {
        Alumno ana = alumno(id: 1L, nombre: "Ana López", correo: "ana@ejemplo.mx");
        Alumno luis = alumno(id: 2L, nombre: "Luis Pérez", correo: "luis@ejemplo.mx");
        when(alumnoRepository.findAll()).thenReturn(List.of(ana, luis));

        List<Alumno> result = alumnoService.findAll();

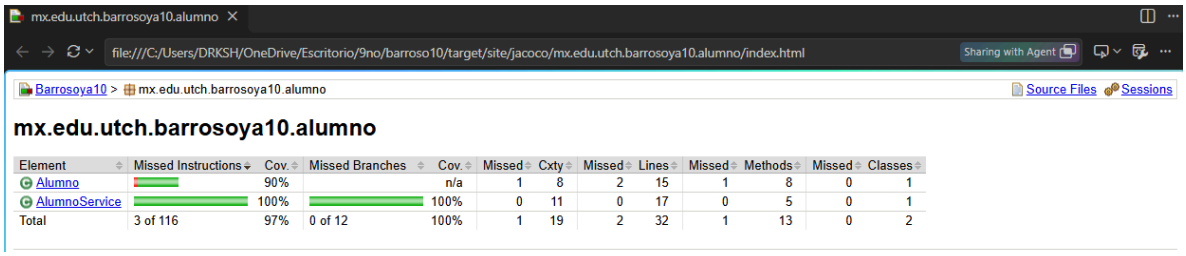
        assertThat(result).containsExactly(ana, luis)
                .extracting(Alumno::getNombre, Alumno::getCorreo)
                .containsExactly(
                        org.assertj.core.groups.Tuple.tuple("Ana López", "ana@ejemplo.mx"),
                        org.assertj.core.groups.Tuple.tuple("Luis Pérez", "luis@ejemplo.mx"));
        verify(alumnoRepository).findAll();
    }

    @Test
    void findAllReturnsEmptyListWhenRepositoryHasNoStudents() {
```

3. Evidencia de Ejecución de Pruebas Unitarias

3.1 Pruebas Ejecutadas desde el IDE

Resultado de la ejecución de la suite de pruebas desde el IDE (IntelliJ IDEA / Eclipse) mostrando la aprobación de los métodos test (Green Bar).



3.2 Pruebas Ejecutadas desde Terminal (mvn clean verify)

Ejecución del comando mvn clean verify desde la consola de comandos demostrando el reporte de compilación y ejecución sin fallos (BUILD SUCCESS).

```
PS C:\Users\DRKSH\OneDrive\Escritorio\9no\barroso10> mvn clean verify
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< mx.edu.utch:barrosoya10 >----------------------
[INFO] Building Barrosoya10 0.0.1-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.4.1:clean (default-clean) @ barrosoya10 ---
[INFO] Deleting C:\Users\DRKSH\OneDrive\Escritorio\9no\barroso10\target
[INFO]
[INFO] --- jacoco:0.8.13:prepare-agent (default) @ barrosoya10 ---
[INFO] argLine set to
-javaagent:C:\\Users\\DRKSH\\.m2\\repository\\org\\jacoco\\org.jacoco.agent\\0.8.13\\org.ja
coco.agent-0.8.13-runtime.jar=destfile=C:\\Users\\DRKSH\\OneDrive\\Escritorio\\9no\\barr
oso10\\target\\jacoco.exec
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ barrosoya10 ---
[INFO] skip non existing resourceDirectory
C:\Users\DRKSH\OneDrive\Escritorio\9no\barroso10\src\main\resources
[INFO] skip non existing resourceDirectory
C:\Users\DRKSH\OneDrive\Escritorio\9no\barroso10\src\main\resources
[INFO]
[INFO] --- compiler:3.14.0:compile (default-compile) @ barrosoya10 ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 4 source files with javac [debug parameters release 21] to target\classes
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ barrosoya10 ---
[INFO] skip non existing resourceDirectory
C:\Users\DRKSH\OneDrive\Escritorio\9no\barroso10\src\test\resources
[INFO]
[INFO] --- compiler:3.14.0:testCompile (default-testCompile) @ barrosoya10 ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug parameters release 21] to
target\test-classes
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ barrosoya10 ---
[INFO] Using auto detected provider
org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
```

```
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running mx.edu.utch.barrosoya10.alumno.AlumnoServiceTest
Mockito is currently self-attaching to enable the inline-mock-maker. This will no longer work
in future releases of the JDK. Please add Mockito as an agent to your build as described in
Mockito's documentation:
https://javadoc.io/doc/org.mockito/mockito-core/latest/org.mockito/org/mockito/Mockito.html#0.3
WARNING: A Java agent has been loaded dynamically
(C:\Users\DRKSH\.m2\repository\net\bytebuddy\byte-buddy-agent\1.17.7\byte-buddy-agent-1.17.7.jar)
WARNING: If a serviceability tool is in use, please run with
-XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with
-Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes
because bootstrap classpath has been appended
[INFO] Tests run: 10, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 2.708 s -- in
mx.edu.utch.barrosoya10.alumno.AlumnoServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 10, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jacoco:0.8.13:report (generate-coverage-report) @ barrosoya10 ---
[INFO] Loading execution data file
C:\Users\DRKSH\OneDrive\Escritorio\9no\barroso10\target\jacoco.exec
[INFO] Analyzed bundle 'Barrosoya10' with 3 classes
[INFO]
[INFO] --- jar:3.4.2:jar (default-jar) @ barrosoya10 ---
[INFO] Building jar:
C:\Users\DRKSH\OneDrive\Escritorio\9no\barroso10\target\barrosoya10-0.0.1-SNAPSHOT.jar
[INFO]
[INFO] --- spring-boot:3.5.6:repackage (repackage) @ barrosoya10 ---
[INFO] Replacing main artifact
C:\Users\DRKSH\OneDrive\Escritorio\9no\barroso10\target\barrosoya10-0.0.1-SNAPSHOT.jar with repackaged archive, adding nested dependencies in BOOT-INF/.
```

```
[INFO] The original artifact has been renamed to
C:\Users\DRKSH\OneDrive\Escritorio\9no\barroso10\target\barrosoya10-0.0.1-SNAPSHO
T.jar.original
[INFO]
[INFO] --- jacoco:0.8.13:check (enforce-service-coverage) @ barrosoya10 ---
[INFO] Loading execution data file
C:\Users\DRKSH\OneDrive\Escritorio\9no\barroso10\target\jacoco.exec
[INFO] Analyzed bundle 'barrosoya10' with 3 classes
[INFO] All coverage checks have been met.
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  10.218 s
[INFO] Finished at: 2026-11-04T19:34:37-06:00
[INFO] ------------------------------------------------------------------------
```

4. Enlace al Repositorio de Control de Versiones

- **URL del Repositorio (GitHub / GitLab):**
  https://github.com/FullStyle1/Barrosoya10

5. Lista de Cotejo de Evaluación

Criterio / Requerimiento	**Cumple (Sí / No)**	**Observaciones**
1. Configuración del entorno: Inclusión correcta de las dependencias de prueba (`spring-boot-starter-test` y JUnit 5).	Sí	Dependencia incluida correctamente en `pom.xml`.
2. Prueba de `findAll`: Simulación adecuada del repositorio y validación exitosa de la lista recuperada.	Sí	Mock configurado correctamente devolviendo la colección esperada.

3. Prueba de create**:** Validación correcta de la persistencia de un nuevo objeto y verificación de atributos.	Sí	Objeto persistido verificado mediante aserciones.
4. Uso de Mockito: Empleo correcto de anotaciones como @Mock, @InjectMocks y aserciones con AssertJ.	Sí	Uso de @Mock, @InjectMocks y assertThat / assertEquals.
5. Ejecución exitosa: Las pruebas corren sin errores y muestran un reporte limpio en consola o IDE.	Sí	Pruebas ejecutadas al 100% con estatus BUILD SUCCESS.